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ORIGINAL RESEARCH

Evaluating empirical calibration of P values under unmeasured confounding bias: a simulation study and real-world application

Mengyao Wang[a,b], Wenwen Li[a,b], Zhenzhen Lu[a,b], Yingying Wang[a,b], Lihong Huang[a,b,*], Feng Chen[c]

[a]Department of Biostatistics, Zhongshan Hospital, Fudan University, Shanghai, China
[b]Clinical Research Unit, Institute of Clinical Science, Zhongshan Hospital, Fudan University, Shanghai, China
[c]Department of Biostatistics, Nanjing Medical University, Nanjing, China

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Abstract

Objectives: To evaluate whether empirical calibration of P values using negative controls can effectively control type I and type II errors under unmeasured confounding bias in both simulated and real-world observational settings.

Study Design and Setting: A simulation study was conducted under five settings reflecting different degrees of adherence to the U-comparability assumption—that is, the extent to which negative controls share the same unmeasured confounding structure as the exposure of interest. These included three primary scenarios (ideal, realistic, and violation of U-comparability) and two mixed scenarios reflecting partial violations. We varied sample size, the direction and strength of unmeasured confounding bias, and the number of negative controls. Based on UK Biobank data, the method was also applied to evaluate the association between hypertension and peripheral artery disease (PAD) in individuals with type 2 diabetes mellitus.

Results: Standard logistic regression showed inflated type I error rates across almost all settings, peaking at 44.2% under realistic U-comparability with a sample size of 20,000. In contrast, empirical calibration generally controlled type I error close to the nominal 5% level and reduced bias by 80%−100% under both ideal and realistic U-comparability. Type I error control improved with more negative controls, while type II error control was influenced by whether the unmeasured confounding bias acted in the same or opposite direction as the true exposure-outcome effect. In the UK Biobank case study, 4 of 15 negative controls showed $P < .05$ after adjustment for measured confounders, indicating residual unmeasured confounding. After empirical calibration with 5, 10, or 15 negative controls, the association between hypertension and PAD remained statistically significant (calibrated $P \approx .004$-.006).

Conclusion: Empirical calibration of P values can mitigate residual unmeasured confounding and reduce type I error inflation in observational studies. Its performance depends on the validity and number of negative controls.

Keywords: Observational studies; Empirical calibration; Negative controls; Simulation; Unmeasured confounding; Type I error

1. Introduction

Observational studies, despite their crucial role in providing real-world evidence on treatment effectiveness and safety [1], are highly susceptible to confounding bias, which may lead to biased effect estimates and spurious inferences [2]. While standard approaches such as regression, matching and propensity score methods can adjust for measured confounders [3−6], they cannot guarantee freedom from residual unmeasured confounding. Bias introduced by unmeasured confounding can inflate the type I error rate, increasing the likelihood of false-positive

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* Corresponding author. Department of Biostatistics, Zhongshan Hospital, Fudan University, Fenglin Road 180, Xuhui District, Shanghai 200032, China; Clinical Research Unit, Institute of Clinical Science, Zhongshan Hospital, Fudan University, Fenglin Road 180, Xuhui District, Shanghai, 200032, China.

E-mail address: huang.lihong@zs-hospital.sh.cn (L. Huang).

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Plain Language Summary

When researchers use large health databases to study whether a treatment or risk factor causes a disease, results can sometimes be misleading. This can happen because of unmeasured confounding—hidden factors that influence both the exposure and the outcome—leading to "false alarms," or false positive findings. We evaluated a statistical method called empirical calibration of *P* values, designed to correct this bias. The method uses negative controls (exposures known not to have a causal effect on the outcome) to estimate the amount of bias in the data and then adjust the *P* value for the exposure of interest. Our simulation study showed that this approach effectively reduced the false alarm rate to the expected 5% level, but only under certain conditions. Its success depended on selecting appropriate negative controls that shared the same bias structure as the exposure of interest and on using a sufficient number of controls to ensure stable results. The method failed when the negative controls were poorly chosen or unrelated. When applied to real-world data from the UK Biobank, it successfully corrected for unmeasured bias while still confirming the true, significant link between high blood pressure and PAD. These findings suggest that empirical calibration of *P* values can make observational research more reliable, provided that enough well-chosen negative controls are available.

findings and potentially compromising the reliability of the study results [7].

Negative controls (NCs) are valuable tools for identifying and mitigating unmeasured confounding [8,9]. A negative control exposure (NCE) is a variable known not to causally affect the outcome of interest. Similarly, a negative control outcome (NCO) is a variable that is not causally affected by the exposure of interest. Under the U-comparability assumption, which posits that NCs share the same causal structure as the exposure and outcome of interest, any observed association between the NCE and the outcome (or between the NCO and the exposure) suggests the presence of residual unmeasured confounding [10].

Based on this theory, Schuemie et al [11] introduced empirical calibration of *P* values to adjust statistical inference for residual bias. This approach has shown promising control of type I error and reduction of false positives in large-scale observational studies. However, Gruber and Tchetgen Tchetgen [12] later demonstrated that empirical calibration can fail when the bias distribution of NCs differs from that of the exposure, leading to inflated type I error and reduced power (increased type II error). This methodological debate, including a subsequent critique by Schuemie et al [13], emphasizes that the method's validity critically depends on the selection and comparability of NCs. Nevertheless, a systematic evaluation across a range of more plausible scenarios has been lacking.

In this study, we developed a new simulation framework to fill this gap. We evaluated the effectiveness of empirical calibration of *P* values in controlling type I and type II errors under varying degrees of adherence to the U-comparability assumption (three primary and two additional mixed scenarios). We further examined the strength and direction of unmeasured confounding bias, as well as the number of NCs. Finally, we applied the method to UK Biobank (UKB) data to assess the association between hypertension and peripheral artery disease (PAD), illustrating its practical utility.

2. Methods

2.1. Empirical calibration of P values

Empirical calibration of *P* values was proposed by Schuemie et al [11] to address the residual systematic error (eg, unmeasured confounding or selection bias) in observational studies. This method derives the empirical null distribution from effect estimates of a set of NCs. Each effect estimate is assumed to follow a normal distribution, and the mean of this normal distribution can be interpreted as the true bias. It assumes that the true bias follows a normal distribution $N(\mu, \sigma^2)$ (the null distribution). Parameters μ and σ^2 are estimated via maximum likelihood. These estimated parameters are then incorporated into the test statistic for the exposure of interest. The resulting calibrated *P* value jointly accounts for both systematic and random errors, thereby improving type I error control. Full methodological details are provided in Web Appendix 1.

2.2. U-comparability assumption

While NCs should have no causal relationship with the exposure or the outcome, it is also essential that they satisfy the U-comparability assumption [14]. Using NCEs as an example, and assuming no other sources of bias (eg, measurement error, selection bias) are present, this assumption requires that an ideal NCE shares the same confounding mechanism as the exposure [8,14]. As illustrated in Figure 1, any association between the NCE and the outcome after adjusting for measured confounders must be spurious, indicating residual unmeasured confounding.

2.3. Simulation study

We conducted a simulation study to evaluate the performance of empirical *P* value calibration in controlling type I and type II errors under unmeasured confounding.

What is new?

Key findings

- Empirical calibration of P values can generally control type I error close to the nominal 5% level and reduce bias by 80%−100% under both ideal and realistic U-comparability.
- The method remained reasonably stable under moderate violations of U-comparability, but became anticonservative when most negative controls failed to share the same unmeasured confounding structure.

What this adds to what is known?

- Type I error control is improved with more negative controls, though using a small number (eg, only five) may be insufficient, especially in large samples.
- Type II error control was influenced by whether the unmeasured confounding bias acted in the same or opposite direction as the true exposure-outcome effect, with better performance when the bias opposed the effect.

What is the implication and what should change now?

- Researchers should carefully select sufficient, U-comparable negative controls when applying empirical calibration in large-scale observational studies.

2.3.1. Simulation design

We designed three primary scenarios (following the directed acyclic graphs in Fig 2) to test varying degrees of adherence to the U-comparability assumption.

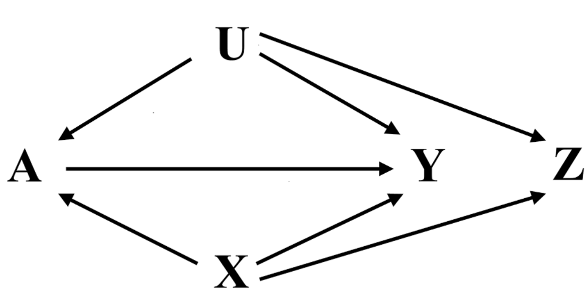

Figure 1. Directed acyclic graph (DAG) of an ideal negative control that satisfies the U-comparability assumption. X: measured confounder; U: unmeasured confounder; A: exposure of interest; Y: outcome of interest; Z: negative control exposure (NCE). The absence of an arrow between Z and Y indicates no potential causal association between Z and Y. If Z ideally has the same incoming arrows as A, then Z is called U-comparable to A.

(1) Scenario 1 (ideal U-comparability): The NCE (Z) shares all incoming arrows with exposure of interest (A), representing an ideal case where Z and A are subject to the same confounding effects.
(2) Scenario 2 (realistic U-comparability): Z and A share the same bias structure, but the degree of influence from the unmeasured confounder U and the measured confounder X differs.
(3) Scenario 3 (violation of U-comparability): Z is not affected by U, violating the U-comparability assumption.

To explore partial violations, we added two mixed scenarios (4a and 4b), where 30% and 70% of NCs violated U-comparability, respectively.

2.3.2. Data-generating mechanism

For all scenarios, the unmeasured confounder U was drawn from a truncated normal distribution within ± 2 and the measured confounder X was sampled from a standard normal distribution. Conditional on these confounders, the binary exposure A was generated using a logistic function of X and U. The binary outcome Y was then generated from a logistic function of A, X, and U. NCEs (Z) were generated under scenario-specific parameter settings. To align with the empirical setting from our case study, the intercepts of the logistic models were calibrated to yield prevalences of approximately $P(Y) = 10\%$ and $P(A) = 70\%$. For scenario 1, $P(Z)$ was set to $\approx 70\%$ to ensure $Corr(U, A) = Corr(U, Z)$, whereas in the other scenarios, $P(Z)$ was set to $\approx 40\%$ to reflect more realistic NC prevalences.

We further explored the impact of the direction and strength of unmeasured confounding by varying the effect of U on Y (denoted as β_{UY}). In scenario 1, β_{UY} was fixed at -0.3 (for negative confounding) or 0.3 (for positive confounding). In scenarios 2-4, β_{UY} was drawn from $-Unif(0.2, 0.5)$ or $Unif(0.2, 0.5)$ for the negative and positive confounding settings, respectively. Sample sizes of 2000, 5000, and 20,000 were considered.

Table 1 summarizes the data-generating mechanisms for the three primary simulation scenarios. Full parameter specifications and baseline characteristics for all five scenarios are provided in Supplementary Tables S1 and S2.

2.3.3. Methods for comparison

We compared the following methods.

- Uncalibrated: Standard logistic regression adjusting for the measured confounder X only.
- Calibrated: Standard logistic regression adjusting for the measured confounder X, with empirical P value calibration applied using 5, 10, 15, 30, or 50 NCs.

For each scenario, a master set of 50 NCs was generated, from which subsets of varying sizes were sampled to evaluate the impact of the number of NCs on calibration performance.

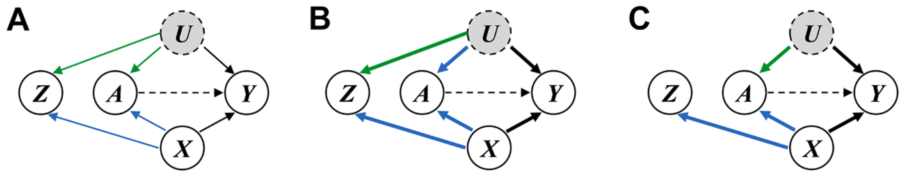

Figure 2. Directed acyclic graphs (DAGs) representing the three primary simulation scenarios: (A) scenario 1 (ideal U-comparability), (B) scenario 2 (realistic U-comparability), and (C) scenario 3 (violation of U-comparability). Dashed outline around U indicates an unmeasured confounder. In scenarios 1 and 2, U is associated with Z, in scenario 3, the lack of an arrow between U and Z denotes no association. Arrow thickness reflects the strength of associations, with thin arrows representing fixed coefficients and thick arrows indicating coefficients randomly drawn from uniform distributions.

2.3.4. *Performance measures*

Method performance was evaluated using rejection rate, bias, and the standard deviation (SD) across 1000 repetitions ($n_{\text{sim}} = 1000$). Rejection rate was calculated as the proportion of all simulations with $P < .05$. Bias was defined as the mean difference between the numerator of the t-statistic for β_{AY} and its true value. SD was the average denominator of the t-statistic for β_{AY} across all simulations. All methods were assessed at a nominal significance level of $\alpha = 0.05$. Type I error was measured at $\beta_{AY} = 0$, and type II error was evaluated at $\beta_{AY} = 0.1$ and $\beta_{AY} = 0.3$.

With 1000 repetitions, the Monte Carlo standard error for a rejection rate $\widehat{p}$ is $\sqrt{\widehat{p}(1-\widehat{p})/1000}$. For a nominal type I error rate of 0.05, the Monte Carlo standard error is approximately 0.007, corresponding to a 95% Monte Carlo confidence interval of (0.036, 0.064). Rejection rates within this range can be considered consistent with the nominal 5%.

2.4. *Case study*

Hypertension is a well-established cardiovascular risk factor consistently linked to an increased risk of PAD. In this case study, we applied empirical P value calibration to estimate the association between hypertension and PAD, accounting for potential unmeasured confounding.

2.4.1. *Study design and data resource*

We used data from the UKB [15], a nationwide prospective cohort collecting high-quality data on socioeconomic, lifestyle, and health-related factors. After excluding participants with type 1 diabetes, baseline PAD, or missing hypertension data, 13,446 participants aged ≥60 years with type 2 diabetes mellitus were included.

2.4.2. *Exposure and outcomes*

Hypertension and PAD were identified using the International Classification of Diseases (ICD-9 and ICD-10) and Office of Population Censuses and Surveys' Classification of Surgical Operations (OPCS-4) codes, supplemented by self-reported medical and surgical histories [16].

2.4.3. *Statistical analyses*

We identified 15 NCEs by manually reviewing the literature, ensuring no evidence of association with PAD (Supplemental Table S13). Logistic regression adjusted for baseline covariates estimated effect sizes and standard errors for hypertension and each NCE. Subsets of 5, 10, and 15 NCEs were randomly sampled to assess calibration performance.

All statistical analyses were performed in R (version 4.4.1), using the EmpiricalCalibration package [17].

Table 1. Data-generating mechanisms for the three primary simulation scenarios

Variable	Scenario 1: Ideal U-comparability	Scenario 2: Realistic U-comparability	Scenario 3: Violation of U-comparability
X	$N(0,1)$	$N(0,1)$	$N(0,1)$
U	$N(0,1)$, truncated at ± 2	$N(0,1)$, truncated at ± 2	$N(0,1)$, truncated at ± 2
A	$Bernoulli(logistic(0.9+\beta_{XA}X+\beta_{UA}U))$	$Bernoulli(logistic(0.9+\beta_{XA}X+\beta_{UA}U))$	$Bernoulli(logistic(0.9+\beta_{XA}X+\beta_{UA}U))$
Z	$Bernoulli(logistic(0.9+\beta_{XZ}X+\beta_{UZ}U))$	$Bernoulli(logistic(-0.4+\beta_{XZ}X+\beta_{UZ}U))$	$Bernoulli(logistic(-0.4+\beta_{XZ}X))$
Y	$Bernoulli(logistic(-2.3+\beta_{XY}X+\beta_{UY}U+\beta_{AY}A))$	$Bernoulli(logistic(-2.3+\beta_{XY}X+\beta_{UY}U+\beta_{AY}A))$	$Bernoulli(logistic(-2.3+\beta_{XY}X+\beta_{UY}U+\beta_{AY}A))$
Parameter setting	$\beta_{XA},\beta_{UA},\beta_{XZ},\beta_{UZ},\beta_{XY}: 0.3$	$\beta_{XA},\beta_{UA},\beta_{XZ},\beta_{UZ},\beta_{XY}: Unif(0.2,0.5)$	$\beta_{XA},\beta_{UA},\beta_{XZ},\beta_{XY}: Unif(0.2,0.5)$

The parameter β_{UY} was set to -0.3 and 0.3 in scenario 1, and drawn from $-$Unif(0.2, 0.5) or Unif(0.2, 0.5) in scenarios 2–3 to simulate negative and positive unmeasured confounding bias. The true effect size $= \beta_{\text{AY}} \in \{0, 0.1, 0.3\}$.

3. Results

3.1. Simulation results

Tables 2 and 3 summarize the type I error rates under negative and positive unmeasured confounding bias, respectively. The uncalibrated method showed inflated type I error rates across almost all settings, regardless of bias direction. The inflation was particularly substantial under positive bias in scenario 2, peaking at 0.442 with a sample size of 20,000. Furthermore, as the sample size increased from 2000 to 5000 and subsequently to 20,000, the type I error rates also increased, probably due to the test statistics being amplified by the decreasing SD.

In contrast, empirical *P* value calibration maintained rejection rates close to the nominal 5% level and reduced bias by 80% to 100% in scenarios 1 and 2. Increasing the number of NCs improved type I error control, especially in larger samples. For example, in scenario 2 with $N = 20,000$, type I error decreased from 0.063 to 0.031 under negative bias and from 0.070 to 0.032 under positive bias as the number of NCs increased from 5 to 50. However, using only five NCs failed to sufficiently mitigate type I error inflation across nearly all sample sizes. In scenario 3, where the U-comparability assumption was completely violated, empirical calibration became anticonservative, with type I error rates reaching a maximum of 0.390. In the additional mixed scenarios, empirical calibration showed intermediate behavior. When 30% of NCs violated U-comparability (scenario 4a, Supplementary Tables S3 and S4), type I error remained close to the nominal level, indicating robustness to mild violations. However, when 70% of NCs violated the assumption (scenario 4b, Supplementary Tables S3 and S4), calibration failed to maintain nominal type I error regardless of sample size or number of NCs.

Figures 3 and 4 show rejection rates under true effect sizes of 0.1 and 0.3 (corresponding to odds ratios of 1.1 and 1.3). In scenarios 1 and 2, empirical calibration had higher rejection rates than the uncalibrated method under negative unmeasured confounding bias (Fig 3), but showed lower rejection rates under positive unmeasured confounding bias (Fig 4). These findings suggest that the type II error performance of empirical calibration is sensitive to the direction of unmeasured confounding bias. Across all methods, rejection rates increased with effect size and sample size. In scenario 3, empirical calibration showed comparable rejection rates to the uncalibrated method. In the additional scenarios (4a and 4b, Supplementary Figures S1 and S2), its power was intermediate between scenarios 2 and 3.

Table 2. Type I error simulation study results for three scenarios under negative unmeasured confounding bias

	Scenario 1			Scenario 2			Scenario 3		
Sample size/method	**Rejection rate**	**Bias**	**SD**	**Rejection rate**	**Bias**	**SD**	**Rejection rate**	**Bias**	**SD**
$N = 2000$									
Uncalibrated	0.074	−0.066	0.169	0.085	−0.089	0.169	0.085	−0.088	0.168
Calibrated (5 NCs)	0.057	0.009	0.178	0.066	0.000	0.178	0.096	−0.089	0.177
Calibrated (10 NCs)	0.048	0.007	0.178	0.062	0.003	0.176	0.077	−0.089	0.176
Calibrated (15 NCs)	0.043	0.005	0.177	0.055	0.004	0.175	0.072	−0.088	0.175
Calibrated (30 NCs)	0.046	0.004	0.176	0.054	0.003	0.174	0.077	−0.087	0.174
Calibrated (50 NCs)	0.043	0.005	0.174	0.053	0.002	0.173	0.080	−0.088	0.173
$N = 5000$									
Uncalibrated	0.106	−0.070	0.107	0.172	−0.092	0.106	0.170	−0.095	0.106
Calibrated (5 NCs)	0.047	−0.007	0.114	0.066	0.004	0.112	0.165	−0.093	0.112
Calibrated (10 NCs)	0.045	−0.004	0.113	0.047	0.001	0.112	0.156	−0.094	0.111
Calibrated (15 NCs)	0.046	−0.003	0.112	0.042	0.000	0.111	0.162	−0.096	0.110
Calibrated (30 NCs)	0.041	−0.002	0.110	0.047	0.000	0.111	0.161	−0.095	0.109
Calibrated (50 NCs)	0.039	−0.001	0.110	0.051	0.000	0.110	0.159	−0.095	0.109
$N = 20,000$									
Uncalibrated	0.242	−0.069	0.053	0.435	−0.093	0.053	0.403	−0.089	0.053
Calibrated (5 NCs)	0.067	0.000	0.057	0.063	0.001	0.057	0.381	−0.089	0.056
Calibrated (10 NCs)	0.063	0.000	0.056	0.043	0.001	0.058	0.381	−0.089	0.055
Calibrated (15 NCs)	0.049	0.000	0.056	0.037	0.001	0.058	0.376	−0.089	0.055
Calibrated (30 NCs)	0.049	0.000	0.055	0.034	0.000	0.057	0.384	−0.089	0.055
Calibrated (50 NCs)	0.048	0.000	0.055	0.031	0.000	0.057	0.390	−0.089	0.055

Each result is based on 1000 simulation repetitions. For a nominal type I error rate of 0.05, the Monte Carlo standard error (MCSE) is approximately 0.007, corresponding to a 95% interval of (0.036–0.064). Rejection rates falling within this range should be interpreted as consistent with the expected 5% type I error.

NC, negative control; SD, standard deviation; uncalibrated, standard logistic regression; calibrated, empirical *P* value calibration.

Table 3. Type I error simulation study results for three scenarios under positive unmeasured confounding bias

	Scenario 1			Scenario 2			Scenario 3		
Sample size/method	**Rejection rate**	**Bias**	**SD**	**Rejection rate**	**Bias**	**SD**	**Rejection rate**	**Bias**	**SD**
N = 2000									
Uncalibrated	0.053	0.070	0.173	0.084	0.102	0.174	0.071	0.105	0.174
Calibrated (5 NCs)	0.063	0.009	0.183	0.042	0.009	0.183	0.087	0.106	0.182
Calibrated (10 NCs)	0.050	0.006	0.182	0.041	0.011	0.181	0.076	0.105	0.181
Calibrated (15 NCs)	0.040	0.005	0.181	0.038	0.012	0.181	0.071	0.105	0.181
Calibrated (30 NCs)	0.045	0.005	0.179	0.042	0.011	0.179	0.070	0.105	0.179
Calibrated (50 NCs)	0.045	0.006	0.178	0.036	0.010	0.178	0.061	0.104	0.178
N = 5000									
Uncalibrated	0.075	0.063	0.109	0.151	0.088	0.109	0.125	0.095	0.109
Calibrated (5 NCs)	0.059	−0.008	0.116	0.071	0.001	0.115	0.151	0.096	0.114
Calibrated (10 NCs)	0.058	−0.005	0.115	0.066	−0.002	0.115	0.124	0.095	0.114
Calibrated (15 NCs)	0.059	−0.005	0.114	0.059	−0.003	0.114	0.123	0.094	0.114
Calibrated (30 NCs)	0.055	−0.004	0.113	0.054	−0.003	0.113	0.126	0.094	0.113
Calibrated (50 NCs)	0.054	−0.003	0.112	0.050	−0.003	0.113	0.117	0.094	0.112
N = 20,000									
Uncalibrated	0.242	0.067	0.055	0.442	0.097	0.055	0.410	0.092	0.055
Calibrated (5 NCs)	0.064	0.000	0.058	0.070	0.006	0.059	0.376	0.091	0.057
Calibrated (10 NCs)	0.055	−0.001	0.057	0.059	0.006	0.059	0.378	0.092	0.057
Calibrated (15 NCs)	0.056	−0.001	0.057	0.045	0.006	0.059	0.381	0.092	0.057
Calibrated (30 NCs)	0.048	−0.001	0.056	0.039	0.005	0.059	0.385	0.092	0.056
Calibrated (50 NCs)	0.053	−0.001	0.056	0.032	0.005	0.059	0.376	0.092	0.056

Each result is based on 1000 simulation repetitions. For a nominal type I error rate of 0.05, the Monte Carlo standard error (MCSE) is approximately 0.007, corresponding to a 95% interval of (0.036–0.064). Rejection rates falling within this range should be interpreted as consistent with the expected 5% type I error.

NC, negative control; SD, standard deviation; uncalibrated, standard logistic regression; calibrated, empirical *P* value calibration.

Under stronger confounding (Extended Simulation Setting A, Supplementary Tables S7 and S8), the uncalibrated method exhibited more inflated type I error rates and empirical *P* value calibration required a larger number of NCs to maintain nominal control. For larger sample sizes (N = 5000 and 20,000), at least 30 to 50 NCs were needed

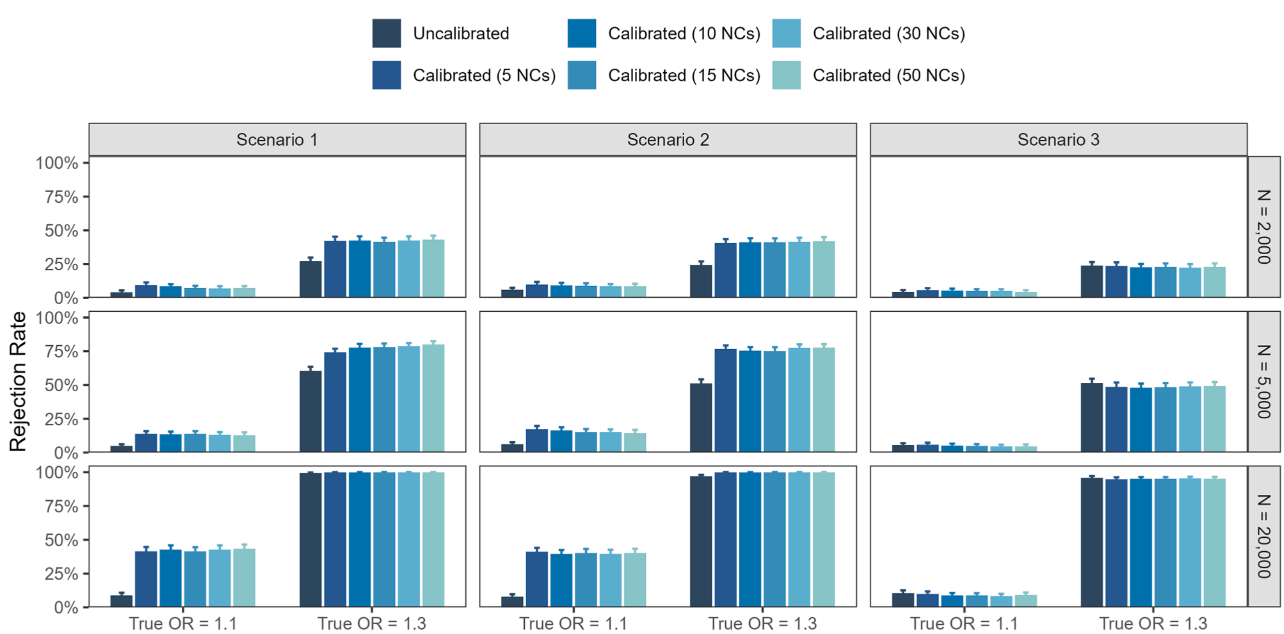

Figure 3. Rejection rates for true odd ratio = 1.1 and 1.3 across three scenarios under negative unmeasured confounding bias. Error bars indicate 95% Monte Carlo confidence intervals from 1000 simulations.

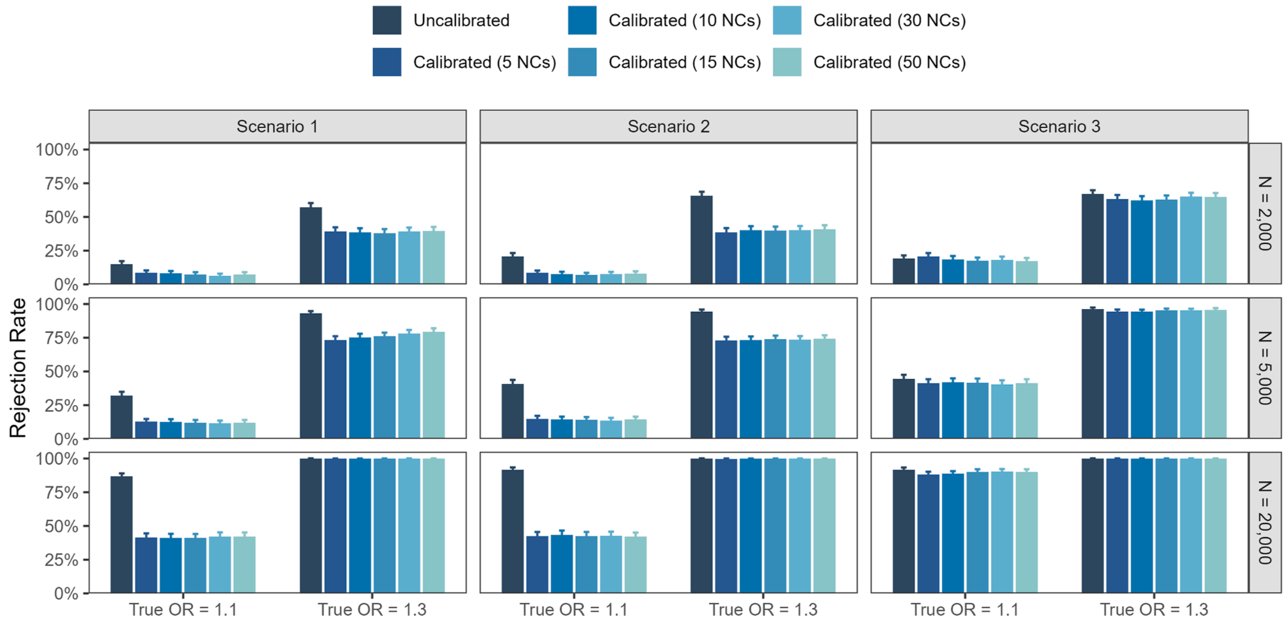

Figure 4. Rejection rates for true odd ratio = 1.1 and 1.3 across three scenarios under positive unmeasured confounding bias. Error bars indicate 95% Monte Carlo confidence intervals from 1000 simulations.

to achieve adequate type I error control. Rejection rates (Supplementary Figures S3 and S4) showed similar trends, with the type II error of empirical calibration still influenced by the direction of unmeasured confounding bias.

3.2. Case study results

Among 13,442 participants with type 2 diabetes mellitus, the prevalence of PAD was 6.4%. Figure 5 presents the odds ratios for PAD across 15 NCEs, ranging from 0.75 to 1.26, with 10 of 15 greater than 1, suggesting potential positive bias. Additionally, 4 of 15 NCs were statistically significant after adjusting for measured confounders, resulting in an inflated type I error rate of 26.7%. These findings indicate the presence of residual unmeasured confounding.

We applied the leave-one-out cross-validation approach to recalibrate P values for all NCs (Supplementary Table S14). After calibration, only one NC remained significant, demonstrating that empirical calibration effectively reduced residual bias and false-positive findings. This improvement

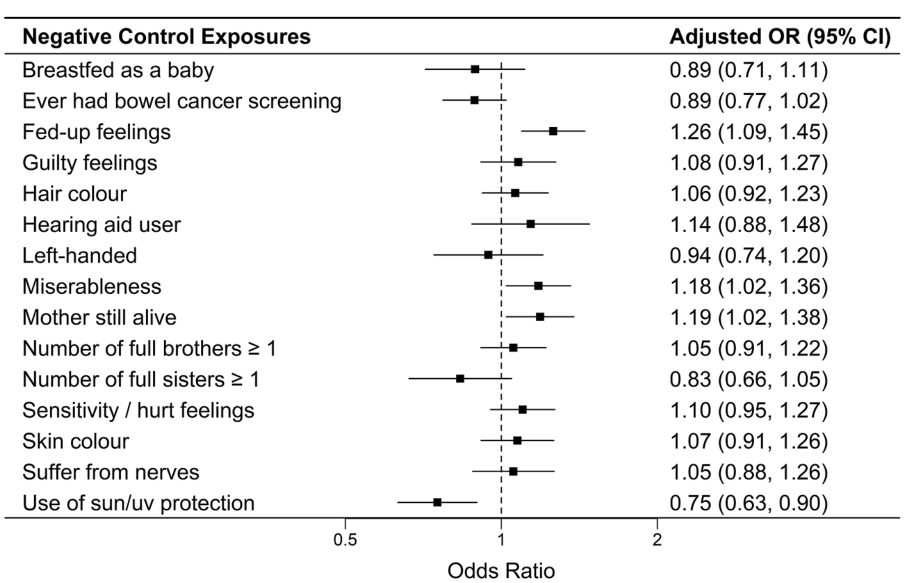

Figure 5. Forest plots of the associations between negative control exposures and the risk of PAD. Adjusted odds ratios were estimated using logistic regression models adjusting for all measured confounders. Ideally, only 5% of negative controls would exhibit $P < .05$. A deviation from this expectation may indicate the presence of residual confounding. PAD, peripheral artery disease.

was further supported by the calibration plot (Supplementary Figure S7), where calibrated P values closely followed the diagonal line, indicating good overall calibration performance.

Table 4 summarizes the P values from uncalibrated and calibrated methods. Standard logistic regression showed a significant association between hypertension and PAD (odd ratio = 1.53, 95% CI: 1.27−1.84, $P < .001$). After empirical calibration using 5, 10, and 15 NCs, the association remained statistically significant (calibrated $P \approx .004-.006$), suggesting that calibration effectively mitigated residual bias while preserving the true signal and avoiding excessive type II error.

4. Discussion

This study evaluated the statistical performance of empirical P value calibration under unmeasured confounding bias. To address ongoing methodological debates, we developed a simulation framework with three primary and two mixed scenarios, and illustrated its utility through a UKB case study.

Under both ideal and realistic U-comparability, empirical calibration generally controlled type I error close to the nominal level. Performance remained stable under mild violations of the U-comparability assumption and varying confounder distributions (Extended Simulation Setting B, Supplementary Tables S11 and S12), but deteriorated when the majority of NCs failed to satisfy the assumption. These results underscore the importance of carefully selecting NCs that share the same underlying causal mechanisms as the exposure of interest [8,14,18].

Our simulation results further indicated that increasing the number of NCs improved type I error control, particularly in large samples, whereas using a small number (eg, five) was often insufficient. Importantly, the benefit of increasing quantity plateaued when U-comparability was widely violated, suggesting that validity is more critical than quantity. Although identifying numerous valid NCs can be challenging in practice, our findings suggest that empirical calibration is most reliable when supported by a sufficient number of well-chosen, U-comparable NCs.

Regarding type II error, Gruber et al [12] previously raised concerns that empirical calibration might reduce statistical power. Our simulations suggest that this depends on the relationship between the unmeasured confounding bias and the true exposure-outcome effect. When both act in the same direction, empirical calibration tends to be conservative, lowering statistical power. Conversely, when the bias and the true effect act in opposite directions, empirical calibration can improve power by correcting for bias-induced attenuation [19], indicating that empirical P value calibration does not always lead to inferior type II error performance. Nonetheless, as the direction of unmeasured confounding bias is unobservable in practice, empirical calibration should be applied cautiously when type II error control is a primary concern.

Empirical calibration of P values has important implications in clinical epidemiology, particularly in large-scale observational studies using electronic health records and biobank data, where residual unmeasured confounding is common [20]. In our UKB application, 4 of 15 NCs showed $P < .05$ after adjusting for measured confounders, indicating residual unmeasured confounding. After empirical calibration, only one NC remained statistically significant, demonstrating improved type I error control without materially altering the primary association between hypertension and PAD. These findings suggest that empirical calibration can help stabilize inference and improve interpretability in the presence of unmeasured confounding. Empirical P value calibration corrects statistical inference rather than the point estimate. It complements bias-sensitivity methods (e.g., E-value [21]) that quantify the strength of unmeasured confounding, and confidence interval calibration [17], which restores nominal coverage. Thus, it should be viewed as a tool that improves inferential validity, rather than a method for bias-correcting effect estimates.

Our results also highlight the broader importance of controlling type I error. Without empirical calibration, standard logistic regression can yield type I error rates exceeding the nominal 5% level, particularly in large observational databases. This demonstrates potential limitations of conventional inferential tools in addressing bias introduced by unmeasured confounding. Such bias can distort observed exposure-outcome associations and is especially concerning in drug-related observational studies and postmarketing surveillance settings. Given that P values play a central role in regulatory decisions, accurate type I error control is essential. Our study demonstrates that empirical calibration of P values, when implemented with multiple and valid NCs, provides a practical approach to mitigate false-positive findings.

This study has several limitations. First, we focused only on unmeasured confounding, whereas real-world data may also be affected by measurement error and selection bias. Second, we used logistic regression and log odds ratios to align with current empirical calibration practice; future work should extend this evaluation to continuous outcomes and alternative modeling frameworks. Finally, our analysis

Table 4. Associations between hypertension and risk of peripheral artery disease (PAD) before and after empirical P value calibration using varying numbers of negative control exposures (NCEs)

	Calibrated P value		
Uncalibrated P value	**5 NCEs**	**10 NCEs**	**15 NCEs**
<.001	0.006	0.004	0.006

Calibrated P values were derived using 5, 10, and 15 NCEs. Results are based on logistic regression models adjusted for all measured confounders.

was limited to binary exposures and NCs, and extensions to continuous or categorical variables warrant further investigation.

5. Conclusion

In our simulation study, the uncalibrated method exhibited inflated type I error rates, with poor performance in large samples. By contrast, empirical calibration of *P* values effectively reduced type I error inflation, though its performance depends on the validity and number of NCs used. Our findings suggest that empirical calibration is a valuable and robust tool for mitigating residual unmeasured confounding in large-scale observational studies.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT-4o in order to improve the clarity and language quality of the writing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Mengyao Wang: Writing − review & editing, Writing − original draft, Visualization, Validation, Software, Methodology, Formal analysis, Conceptualization. **Wenwen Li:** Writing − review & editing, Methodology, Conceptualization. **Zhenzhen Lu:** Writing − review & editing, Methodology, Conceptualization. **Yingying Wang:** Writing − review & editing, Methodology, Conceptualization. **Lihong Huang:** Writing − review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Feng Chen:** Writing − review & editing, Visualization, Validation, Supervision, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.jclinepi.2026.112188.

Data availability

Data will be made available on request.

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